

Analysis of S2 star orbital residuals using a spacetime refraction model (WRM)

1. Assumptions for the analysis: Geometric parameters of the universe

This analysis uses only real numbers based on the following observational facts, eliminating arbitrary adjustment coefficients based on the Spacetime Refraction Model (WRM).

- **Observation distance** L : 2.6×10^4 light years
- **The entire universe** R_{total} : 1.38×10^{10} light years
- **Dipole azimuth** θ : 93.2°
- **Cosine value at perihelion**: $\cos(93.2^\circ) \approx -0.0558$
- **Standard advance angle** $\Delta\phi_{GR}$: about $12' = 720''$

2. Mathematical derivation: Calculation of amplification factor using refractive gradient

This method derives the deviation (fractional value) of observed values due to the structural gradient of the universe.

$$A_{worm} = \exp\left(\frac{L}{R_{total}} \cdot \cos\theta\right)$$
$$A_{worm} = \exp\left(\frac{26,000}{13,800,000,000} \cdot (-0.0558)\right)$$
$$A_{worm} \approx \exp(-1.05 \times 10^{-7}) \approx 1 - 0.000000105$$

This ratio is applied to the orbital angle and converted to arcseconds.

$$\text{sidedifferenceReasontheoryvalue} = \Delta\phi_{GR} \times (A_{worm} - 1)$$
$$\text{sidedifferenceReasontheoryvalue} \approx 720'' \times (-1.05 \times 10^{-7})$$
$$\text{sidedifferenceReasontheoryvalue} \approx -0.0756 \text{ MiRisecondscorner (mas)}$$

3. Clear indication of numerical agreement

Comparison items	numerical values	physical meaning
Residuals (RMS) of the GRAVITY paper	0.05 to 0.1 mas	Actual observed values that could not be explained by the standard model (GR) and have been noise-processed.
Standard score based on WRM theory	0.0756 mas	The inevitable correction value derived from the structural gradient of a cosmic resonator.

Conclusion: The fractional values of 0.1 mas or less that modern astronomy considers to be "noise due to observational limits" are, in WRM, related to the geometric arrangement of the universe (L, R_{total}, θ It matches the number derived solely from).

4. References

- **Abuter, R., et al. (GRAVITY Collaboration).** (2020). "Detection of the Schwarzschild precession in the orbit of the star S2 near the Galactic center." *A&A* 636, L5.
- **Planck Collaboration.** (2018). "Planck 2018 results. I. Cosmic microwave background dipole."
- "Technical Notes on the Apparent Hyperfast Motion of Quasars" by **Hiroshi Haraguchi (Effective Refractive Index) n_{eff} (Definition)**

Spacetime refraction model (WRM) analysis: S55 star (S0-102)

1. Input real number parameters

- **Observation distance** L : 2.6×10^4 light years
- **The entire universe** R_{total} : 1.38×10^{10} light years
- **Azimuth** θ : 93.2°
- **Cosine value**: $\cos(93.2^\circ) \approx -0.0558$
- **S55 specific advance angle** $\Delta\phi_{GR}$: about $780''$ (Theoretical value of Schwarzschild propagation per orbit)

2. The process of expanding arithmetic expressions

amplification factor A_{worm} Calculation of:

$$A_{worm} = \exp\left(\frac{26,000}{13,800,000,000} \times (-0.0558)\right)$$
$$A_{worm} = \exp(0.000001884 \times -0.0558)$$
$$A_{worm} = \exp(-1.051 \times 10^{-7}) \approx 0.9999998949$$

Calculation of the deviation (fraction) of angular displacement:

$$\text{sidedifferenceReasontheoryvalue} = 780'' \times (0.9999998949 - 1)$$
$$\text{sidedifferenceReasontheoryvalue} = 780'' \times (-1.051 \times 10^{-7})$$
$$\text{sidedifferenceReasontheoryvalue} = -0.0000819 \text{ secondscorner} \approx -\mathbf{0.0819 \text{ MiRisecon}}$$

3. Agreement with observed values

item	numerical values	basis
Residual (RMS) of S55 observation data	Approximately 0.08~0.12mas	Peissker et al. (2020) Residuals
WRM calculation deviation	Approximately 0.082 mas	Theoretical real numbers based on geometric derivation

Spacetime refraction model (WRM) analysis: S62 star

1. Input real number parameters

- **Observation distance** L : 2.6×10^4 light years
- **The entire universe** R_{total} : 1.38×10^{10} light years
- **Azimuth** θ : 93.2°
- **Cosine value**: $\cos(93.2^\circ) \approx -0.0558$
- **S62 specific advance angle** $\Delta\phi_{GR}$: about $1,100''$ (Huge theoretical trajectory values due to extremely close-range orbits)

2. The process of expanding arithmetic expressions

amplification factor A_{worm} Calculation of:

$$A_{worm} = \exp\left(\frac{26,000}{13,800,000,000} \times (-0.0558)\right) \approx 0.9999998949$$

Calculation of the deviation (fraction) of angular displacement:

sidedifferenceReasontheoryvalue = $1,100'' \times (A_{worm} - 1)$
sidedifferenceReasontheoryvalue = $1,100'' \times (-1.051 \times 10^{-7})$
sidedifferenceReasontheoryvalue = -0.0001156 secondscorner $\approx -\mathbf{0.1156}$ MiRisecorner

3. Numerical agreement

item	numerical values	Physical interpretation
Residual (RMS) of S62 orbital analysis	Approximately 0.10~0.15mas	The measured deviation treated as noise by Peissker et al. (2020).
WRM calculation deviation	Approximately 0.116mas	The inevitable structural errors predicted by the cosmic resonator model.